Book: Measurement and Data Analysis for Engineering and Science by Patrick Dunn

11.5 MATLAB SIDEBAR

The rms can be computed using the MATLAB `norm` command. This expression `norm(x)` equals `sum[abs(x).^2]^1/2`. The "." is present in this expression because of the multiplication of the vector `x` with itself. So, the rms for the discrete signal `x` can be written in MATLAB as

```
rms=norm(x)/sqrt(length(x));
```

So, the rms of the fluctuating component is

$$u_{\text{rms}} = \left\{ \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \left[U(t) - \overline{U}(t) \right]^2 dt \right\}^{1/2}, \qquad \textbf{[11.27]}$$

where

$$\overline{U}(t) = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} U(t)\, dt. \qquad \textbf{[11.28]}$$

By comparing Equations 11.25 and 11.27, it is evident that $U_{\text{rms}} \neq u_{\text{rms}}$.

Example 11.3

STATEMENT

Determine the rms of the ramp function $y(t) = A(t/T)$ in which A is the amplitude and T is the period.

SOLUTION

Because $y(t)$ is a deterministic periodic function, the rms needs to be computed for only one period, from $t = 0$ to $t = T$. Application of the rms equation from Table 11.1 for $y(t)$, which is a continuous signal, yields

$$y_{\text{rms}} = \left[\frac{A^2}{T^2(t_2 - t_1)} \int_{t_1}^{t_2} t^2\, dt \right]^{1/2} = \frac{A^2(t_2^3 - t_1^3)}{3T^2(t_2 - t_1)}.$$

For $t_1 = 0$ and $t_2 = T$, the rms becomes

$$y_{\text{rms}} = \frac{A}{\sqrt{3}}.$$

What is $\overline{y}(t)$? (Answer: $A/2$.) What is the rms of a sine wave of amplitude A? (Answer: $A/\sqrt{2}$.)

11.7 FOURIER SERIES OF A PERIODIC SIGNAL

Before considering the Fourier series, the definition of orthogonality must be examined. The **inner product** (dot product), (x, y), of two real-valued functions $x(t)$ and

$y(t)$ over the interval $a \le t \le b$ is defined as

$$(x, y) = \int_a^b x(t)y(t)\,dt. \qquad [11.29]$$

If $(x, y) = 0$ over that interval, then the functions x and y are **orthogonal** in the interval. If each *distinct* pair of functions in a set of functions is orthogonal, then the set of functions is **mutually orthogonal**.

For example, the set of functions $\sin(2\pi mt/T)$ and $\cos(2\pi mt/T), m = 1, 2, \ldots,$ form one distinct pair and are mutually orthogonal because

$$\int_{-T/2}^{T/2} \sin\left(\frac{2\pi mt}{T}\right)\cos\left(\frac{2\pi nt}{T}\right)dt = 0 \text{ for all } m,\ n. \qquad [11.30]$$

Also, these functions satisfy the other orthogonality relations

$$y(t) = \int_{-T/2}^{T/2} \cos\left(\frac{2\pi mt}{T}\right)\cos\left(\frac{2\pi nt}{T}\right)dt = \begin{cases} 0 & m \neq n \\ T & m = n \end{cases} \qquad [11.31]$$

and

$$y(t) = \int_{-T/2}^{T/2} \sin\left(\frac{2\pi mt}{T}\right)\sin\left(\frac{2\pi nt}{T}\right)dt = \begin{cases} 0 & m \neq n \\ T & m = n. \end{cases} \qquad [11.32]$$

Knowing these facts is useful when performing certain integrals, such as those that occur when determining the Fourier coefficients.

Fourier analysis and synthesis, named after Jean-Baptiste-Joseph Fourier (1768–1830), a French mathematician, now can be examined. Fourier showed that the temperature distribution through a body could be represented by a series of harmonically related sinusoids. The mathematical theory for this, however, actually was developed by others [3]. Fourier methods enable complex signals to be approximated in terms of a series of sines and cosines. This is called the trigonometric **Fourier series**. The Fourier trigonometric series that represents a signal of period T can be expressed as

$$y(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty}\left[A_n \cos\left(\frac{2\pi nt}{T}\right) + B_n \sin\left(\frac{2\pi nt}{T}\right)\right], \qquad [11.33]$$

where

$$A_0 = \frac{2}{T}\int_{-T/2}^{T/2} y(t)\,dt, \qquad [11.34]$$

$$A_n = \frac{2}{T}\int_{-T/2}^{T/2} y(t)\cos\left(\frac{2\pi nt}{T}\right)dt \quad n = 1, 2, \ldots, \qquad [11.35]$$

and

$$B_n = \frac{2}{T}\int_{-T/2}^{T/2} y(t)\sin\left(\frac{2\pi nt}{T}\right)dt \quad n = 1, 2, \ldots. \qquad [11.36]$$

The frequencies associated with the sines and cosines are integer multiples (*nth harmonics*) of the fundamental frequency. The *fundamental* or *primary* frequency, the first harmonic, is denoted by $n = 1$, the second harmonic by $n = 2$, the third harmonic by $n = 3$, and so on. A_0 is twice the average of $y(t)$ over one period. A_n and B_n are called the **Fourier coefficients** of the Fourier amplitudes. The expression for A_n can be determined by multiplying both sides of the original series expression for $y(t)$ by $\cos(2\pi nt/T)$, then integrating over one period from $t = -T/2$ to $t = T/2$. The expression for B_n is found similarly but instead by multiplying by $\sin(2\pi nt/T)$. This is called Fourier's trick.

The procedure by which the Fourier amplitudes for any specified $y(t)$ are found is called **Fourier analysis**. Fourier analysis is the analog to a prism that separates white light (a complex signal) into colors (simple periodic sine functions). **Fourier synthesis** is the reverse procedure by which $y(t)$ is constructed from a series of appropriately weighted sines and cosines. The Fourier synthesis of a signal is useful because the amplitude and frequency components of the signal can be identified.

A Fourier series representation of $y(t)$ exists if $y(t)$ satisfies the following Dirichlet conditions:

1. $y(t)$ has a finite number of discontinuities in the period T (it is piecewise differentiable).
2. $y(t)$ has a finite average value.
3. $y(t)$ has a finite number of relative maxima and minima in the period T.

If these conditions are met, then the series converges to $y(t)$ at the values of t where $y(t)$ is continuous and converges to the mean of $y(t^+)$ and $y(t^-)$ at a finite discontinuity. Fortunately, these conditions hold for most situations.

Recall that a periodic function with period T satisfies $y(t+T) = y(t)$ for all t. It follows that if $y(t)$ is an integrable periodic function with a period T, then the integral of $y(t)$ over *any* interval of length T has the same value. Hence, the limits from $-T/2$ to $T/2$ of the Fourier coefficient integrals can be replaced by, for example, from 0 to T or from $-T/4$ to $3T/4$. Changing these limits sometimes simplifies the integration procedure.

The process of arriving at the Fourier coefficients also can be simplified by examining whether the integrands are either even or odd functions. Example even and odd functions are shown in Figure 11.9. If $y(t)$ is an **even function**, where it is symmetric about the y-axis, then $g(x) = g(-x)$. Thus,

$$\int_{-T}^{T} g(x)\,dx = 2\int_{0}^{T} g(x)\,dx. \qquad \textbf{[11.37]}$$

The cosine is an even function. Likewise, if $y(t)$ is an **odd function**, where it is symmetric about the origin, then $g(x) = -g(-x)$. So,

$$\int_{-T}^{T} g(x)\,dx = 0. \qquad \textbf{[11.38]}$$

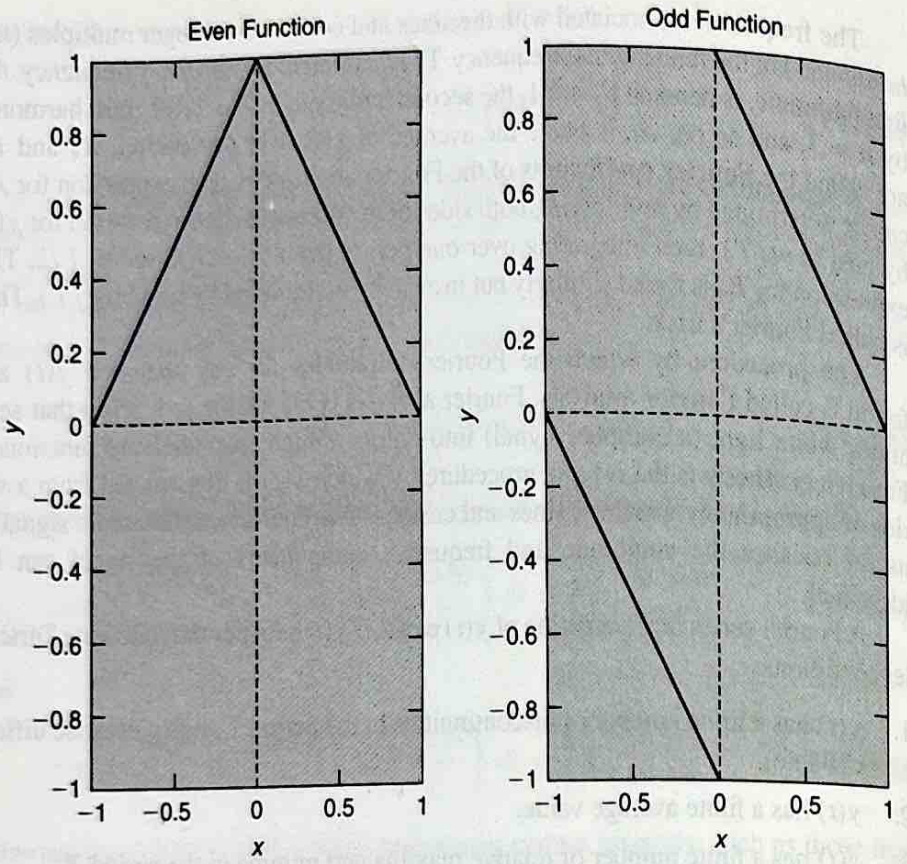


Figure 11.9 Even and odd functions.

The sine is an odd function. Other properties of even and odd functions include the following:

1. The sum, difference, product, or quotient of two even functions are even.
2. The sum or difference of two odd functions are odd.
3. The product or quotient of two odd functions are even.
4. The product or quotient of an even function and an odd function is odd.
5. The sum or difference of an even function and an odd function is neither even nor odd.
6. A general function can be decomposed into a sum of even plus odd functions.

From these properties, Equation 11.33, and the Fourier coefficient equations, it follows that, when $y(t)$ is an even periodic function, $B_n = 0$ and $y(t)$ has the Fourier series

$$y(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty}\left[A_n \cos\left(\frac{2\pi nt}{T}\right)\right]. \qquad \textbf{[11.39]}$$

This is called the *Fourier cosine series*. Further, when $y(t)$ is an odd periodic function, $A_0 = A_n = 0$ and $y(t)$ has the Fourier series

$$y(t) = \sum_{n=1}^{\infty} \left[B_n \sin\left(\frac{2\pi nt}{T}\right) \right]. \qquad [11.40]$$

This is called the *Fourier sine series.*

Example 11.4

STATEMENT

Find the frequency spectrum of the step function

$$y(t) = \begin{cases} -A & -\pi \le t < 0 \\ +A & 0 \le t < \pi \end{cases}$$

SOLUTION

This is an odd function, therefore $A_0 = A_n = 0$.

$$B_n = \frac{2}{T} \int_{-T/2}^{T/2} y(t) \sin\left(\frac{2\pi nt}{T}\right) dt$$

$$\text{note : } \omega = \frac{2\pi}{T} \text{ where } T = 2\pi$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^{0} (-A) \sin(nt)\, dt + \int_{0}^{\pi} (A) \sin(nt)\, dt \right]$$

$$= \frac{1}{\pi} \left\{ \left[\frac{A}{n} \cos(nt) \right]_{-\pi}^{0} - \left[\frac{A}{n} \cos(nt) \right]_{0}^{\pi} \right\}$$

$$= \frac{A}{n\pi} [1 - \cos(-n\pi) - \cos(n\pi) + 1]$$

$$= \frac{2A}{n\pi} [1 - \cos(n\pi)]$$

$$= \frac{4A}{n\pi} \text{ for } n \text{ odd}$$

$$= 0 \text{ for } n \text{ even.}$$

Note that B_n involves only n and constants.

$$\Rightarrow y(t) = \sum_{n=1}^{\infty} B_n \sin(nt)$$

$$= \sum_{n=1}^{\infty} \frac{2A}{n\pi} [1 - \cos(n\pi)] \sin(nt)$$

$$= \frac{4A}{\pi} \sum_{n=1,3,5,\ldots}^{\infty} \left(\frac{1}{n}\right) \sin(2\pi nft)$$

$$= \frac{4A}{\pi} \left[\sin(t) + \frac{1}{3}\sin(3t) + \frac{1}{5}\sin(5t) + \cdots \right],$$

$y(t)$ involves both n and t. The frequencies of each of the sine terms are 1, 3, 5, ... in units of ω (rad/s), or $1/2\pi$, $3/2\pi$, $5/2\pi$, ... in units of f (cycles/s = Hz). Generally, this can be written as $f_n = (2n - 1)f_1$. Likewise, the corresponding amplitudes can be expressed as $A_n = A_1/(2n - 1)$, where $A_1 = 4A/\pi$. $y(t)$ is shown shortly in Figure 11.11 for three different partial sums for $A = 5$.

The contributions of each of the harmonics to the amplitude of the square wave is illustrated in Figure 11.10. The square wave shown along the back plane is the sum of the first 500 harmonics. Only the first five harmonics are given in the figure. The decreasing amplitude and increasing frequency contributions of the next higher harmonic tend to fill in the contributions of the previously summed harmonics such that the resulting wave approaches a square wave.

The partial Fourier series sums of the step function are shown in Figure 11.11 for $N = 1$, 10, and 500. Clearly, the more terms that are included in the sum, the closer the sum approximates the actual step function. Relatively small fluctuations at the end of the step can be seen, especially in the $N = 500$ sum. This is known as the Gibbs phenomenon. The inclusion of more terms in the sum will attenuate these fluctuations but never completely eliminate them.

A plot of amplitude versus frequency can be constructed for a square wave, as presented in Figure 11.12 for the first eight harmonics. This figure illustrates in another way that the Fourier series representation of a square wave consists of multiple frequencies of decreasing amplitudes.

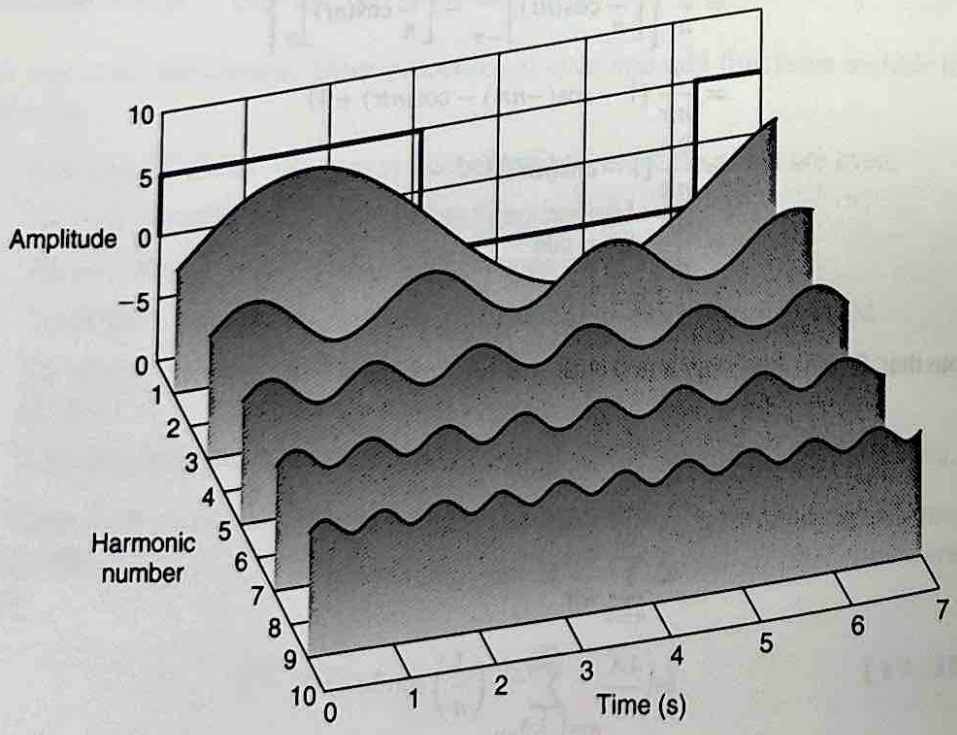


Figure 11.10 Contributions of the first five harmonics to the Fourier series of a step function.

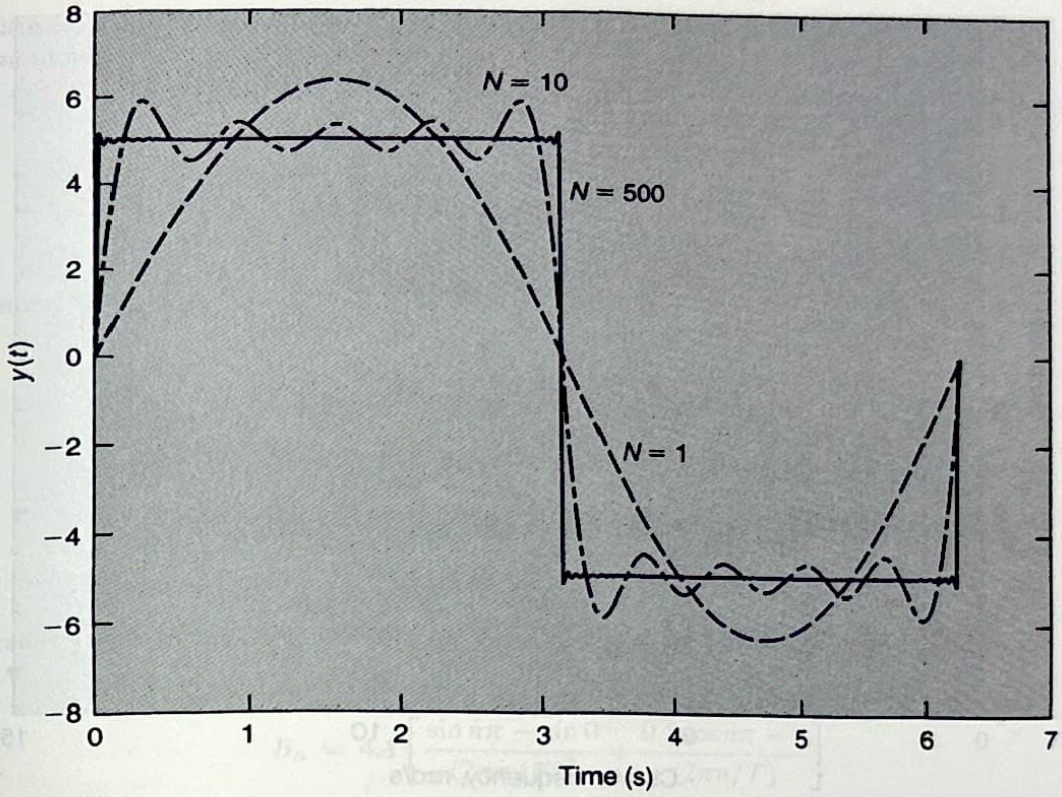


Figure 11.11 Partial Fourier series sums for the step function.

11.6 MATLAB SIDEBAR

The M-file `fsstep.m` was used to construct Figure 11.10. This M-file uses the MATLAB `symsum` command to create the sum of the Fourier series. This is accomplished in part for the sum of 500 harmonics shown in the figure by the syntax

```
syms n t
y=(10/(n*pi))*(1-cos(n*pi))*sin(n*t);
step=symsum(y,n,1.500);
t=0:0.01:7;
ystep=eval(vectorize(step));
```

11.7 MATLAB SIDEBAR

The M-file `sersum3.m` constructs a plot of the Fourier series sum for a user-specified $y(t)$ and three values of N. This was used to construct Figure 11.11. This M-file uses the MATLAB `symsum` command, as described in the previous MATLAB sidebar. It also was used to construct Figure 11.13, which will be discussed later.

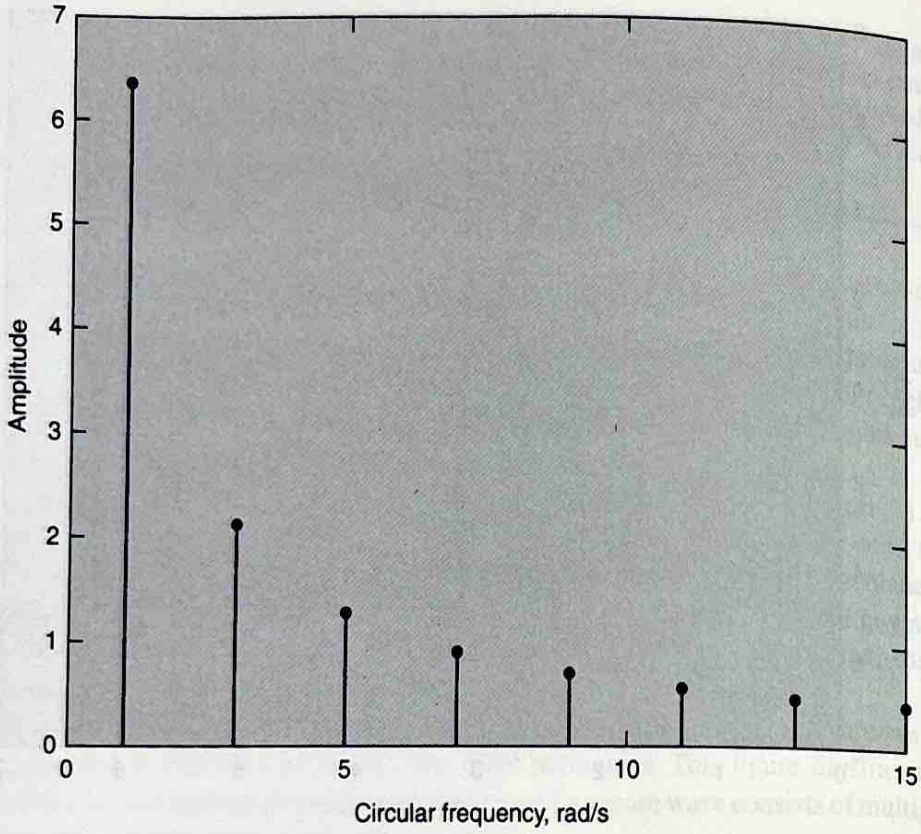


Figure 11.12 Amplitude spectrum for the first eight terms of the step function.

Example 11.5

STATEMENT

Find the frequency spectrum of the ramp function:

$$y(t) = \begin{cases} 2At & 0 \le t < 1/2 \\ 0 & 1/2 \le t < 1 \end{cases}$$

with $T = 1$ s.

SOLUTION

This function is neither even nor odd.

$$A_0 = \frac{2}{T}\int_{-T/2}^{T/2} y(t)\,dt = \frac{2}{1}\left(\int_{-0.5}^{0} 0\,dt + \int_{0}^{0.5} 2At\,dt\right) = 4A\,\frac{t^2}{2}\bigg|_0^{0.5} = \frac{A}{2}$$

$$\begin{aligned} A_n &= \frac{2}{T}\int_{-T/2}^{T/2} y(t)\cos\frac{2\pi nt}{T}\,dt \\ &= 2\int_0^{0.5} 2At\cos\frac{2\pi nt}{T}\,dt \\ &= 4A\left[\frac{\cos(2\pi nt/T)}{(2\pi n/T)^2} + \frac{t\sin(2\pi nt/T)}{(2\pi n/T)}\right]\Bigg|_0^{0.5} \end{aligned}$$

because $\int t \cos mt\, dt = (1/m^2)\cos mt + (t/m)\sin mt$ (from integration by parts).
Continuing,

$$A_n = 4A\left[\frac{\cos n\pi - \cos 0}{(2\pi n/T)^2} + \frac{0.5\sin n\pi - 0}{(2\pi n/T)}\right]$$
$$= (4A)\left(\frac{T^2}{4\pi^2 n^2}\right)(\cos n\pi - 1) = \frac{A}{\pi^2 n^2}(\cos n\pi - 1).$$

Further,

$$B_n = \frac{2}{T}\int_{-T/2}^{T/2} y(t)\sin\frac{2\pi nt}{T}\,dt$$
$$= 2\int_0^{0.5} 2At\sin\frac{2\pi nt}{T}\,dt$$
$$= 4A\left[\frac{\sin(2\pi nt/T)}{(2\pi n/T)^2} - \frac{t\cos(2\pi nt/T)}{(2\pi n/T)}\right]\Bigg|_0^{0.5}$$

because $\int t \sin mt\, dt = (1/m^2)\sin mt - (t/m)\cos mt$ (from integration by parts).
So,

$$B_n = 4A\left[\frac{\sin n\pi - \sin 0}{(2\pi n/T)^2} + \frac{0.5\cos n\pi - 0}{(2\pi n/T)}\right]$$
$$= (4A)\frac{-0.5\cos n\pi}{(2\pi n/T)} = \frac{-A}{\pi n}\cos n\pi.$$

Thus,

$$y(t) = A\left\{\frac{1}{4} + \sum_{n=1}^{\infty}\left[\frac{(\cos n\pi - 1)}{\pi^2 n^2}\cos\frac{2\pi nt}{T} - \frac{\cos n\pi}{\pi n}\sin\frac{2\pi nt}{T}\right]\right\}$$
$$= A\left\{\frac{1}{4} + \sum_{n=1}^{\infty}\frac{1}{n\pi}\left[\frac{(-1+(-1)^n)}{\pi n}\cos\frac{2\pi nt}{T} - (-1)^n\sin\frac{2\pi nt}{T}\right]\right\}.$$

This $y(t)$ is shown (with $A = 1$) for three different partial sums in Figure 11.13.

11.8 EXPONENTIAL FOURIER SERIES

The trigonometric Fourier series can be simplified using complex number notation. Starting with the trigonometric Fourier series

$$y(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty}\left[A_n\cos\left(\frac{2\pi nt}{T}\right) + B_n\sin\left(\frac{2\pi nt}{T}\right)\right], \qquad [11.41]$$

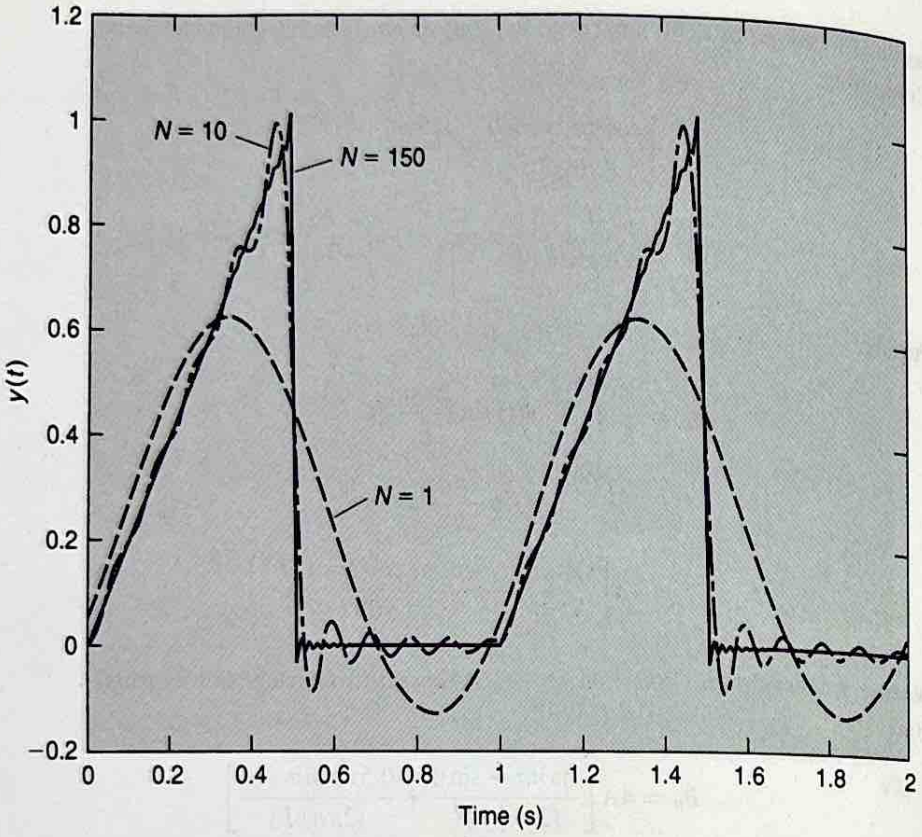


Figure 11.13 Three partial Fourier series sums for a ramp function.

and substituting Equations 11.16 and 11.17 into Equation 11.41 yields

$$y(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty}\left[\frac{A_n}{2}\left(e^{in\omega_0 t} + e^{-in\omega_0 t}\right) + \frac{B_n}{2i}\left(e^{in\omega_0 t} - e^{-in\omega_0 t}\right)\right], \quad \textbf{[11.42]}$$

where $\theta = n\omega_0 t = 2\pi nt/T$. Rearranging the terms in this equation gives

$$y(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty}\left[e^{in\omega_0 t}\left(\frac{A_n}{2} - \frac{iB_n}{2}\right) + e^{-in\omega_0 t}\left(\frac{A_n}{2} + \frac{iB_n}{2}\right)\right], \quad \textbf{[11.43]}$$

noting $1/i = -i$.

Using the definitions

$$C_n \equiv \frac{A_n}{2} - \frac{iB_n}{2}, \quad \textbf{[11.44]}$$

$$C_{-n} \equiv \frac{A_n}{2} + \frac{iB_n}{2}, \quad \textbf{[11.45]}$$

this equation can be simplified as follows:

$$y(t) = \sum_{n=1}^{\infty}\left(C_n e^{in\omega_0 t} + C_{-n}e^{-in\omega_0 t}\right) = \frac{A_0}{2} + \sum_{n=-\infty}^{-1} C_n e^{in\omega_0 t} + \sum_{n=1}^{\infty} C_n e^{in\omega_0 t}. \quad \textbf{[11.46]}$$

Combining the two summations yields

$$y(t) = \sum_{n=-\infty}^{\infty} C_n e^{in\omega_0 t}, \qquad \textbf{[11.47]}$$

where $C_0 = A_0/2$.

The coefficients C_n can be found by multiplying Equation 11.47 by $e^{-in\omega_0 t}$ and then integrating from 0 to T. The integral of the right-hand side equals zero except where $m = n$, which then yields the integral equal to T. Thus,

$$C_m = \frac{1}{T}\int_0^T y(t)e^{-im\omega_0 t}\,dt. \qquad \textbf{[11.48]}$$

What has been done here is noteworthy. An expression (Equation 11.41) involving two coefficients A_n and B_n with sums from $n = 1$ to $n = \infty$ was reduced to a simpler form having one coefficient, C_n, with a sum from $n = -\infty$ to $n = \infty$ (Equation 11.47). This illustrates the power of complex notation.

11.9 SPECTRAL REPRESENTATIONS

Additional information about the process represented by a signal can be gathered by displaying the signal's amplitude components versus their corresponding frequency components. This results in representations of the signal's amplitude, power, and power density versus frequency, which are termed the **amplitude spectrum**, **power spectrum**, and **power density spectrum**, respectively. These spectra can be determined from the Fourier series of the signal.

Consider the time average of the square of $y(t)$,

$$\left\langle [y(t)]^2 \right\rangle \equiv \frac{1}{T}\int_0^T [y(t)]^2\,dt. \qquad \textbf{[11.49]}$$

The square of $y(t)$ in terms of the Fourier complex exponential sums is

$$[y(t)]^2 = \left(\sum_{m=-\infty}^{\infty} C_m e^{im\omega_0 t}\right)\left(\sum_{n=-\infty}^{\infty} C_n e^{in\omega_0 t}\right) = \sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty} C_m C_n e^{i(m+n)\omega_0 t}. \qquad \textbf{[11.50]}$$

If Equation 11.50 is substituted into Equation 11.49 and the integral on the right-hand side is performed, the integral will equal zero unless $m = -n$, where, in that case, it will equal T. This immediately leads to

$$\left\langle [y(t)]^2 \right\rangle = \sum_{n=-\infty}^{\infty} C_n C_{-n} = 2\sum_{n=0}^{\infty}\left(\frac{A_n}{2} - \frac{iB_n}{2}\right)\left(\frac{A_n}{2} + \frac{iB_n}{2}\right)$$

$$= 2\sum_{n=0}^{\infty}\left(\frac{A_n^2}{4} + \frac{B_n^2}{4}\right) = 2\sum_{n=0}^{\infty}|C_n|^2 = \sum_{n=-\infty}^{\infty}|C_n|^2, \qquad \textbf{[11.51]}$$

11.8 MATLAB SIDEBAR

The M-file `tsfft.m` constructs a plot of a signal and its spectra using the MATLAB `fft` and `psd` commands. The M-file is written such that a two-column user-specified data file can be called. The data file contains two columns, time and amplitude. Information about the digital methods used to do this is contained in Chapter 12.

where $y(t)$ is assumed to be real and $\langle [y(t)]^2 \rangle$ is termed the **mean-squared amplitude** or average power. Note that this equation is an approximation to the actual mean squared amplitude whenever the number of summed terms is finite. By comparing Equation 11.50 with the definition of the rms for a discrete signal (see Table 11.1), it can be seen that the power is simply the square of the rms. Recall also that the average power over a time interval equals the product of the total energy expended over that interval and the reciprocal of the time interval.

Equations 11.49 and 11.50 can be combined to yield Parseval's relation for continuous-time periodic signals

$$\frac{1}{T}\int_0^T [y(t)]^2\,dt = \sum_{n=-\infty}^{\infty} |C_n|^2 . \qquad \textbf{[11.52]}$$

This states that the total average power in a periodic signal equals the sum of the average powers of all its harmonic components.

The nth amplitude equals $2\sqrt{|C_n|^2} = \sqrt{A_n^2 + B_n^2}$. The plot of $2\sqrt{|C_n|^2}$ versus the frequency is called the amplitude spectrum of the signal $y(t)$. The ordinate units are those of the amplitude. The plot of $|C_n|^2$ versus the frequency is called the power spectrum of the signal $y(t)$. The ordinate units are those of the amplitude squared. The absolute value squared of the C_n Fourier coefficient, which equals one-quarter of the amplitude squared, gives the amount of power associated with the nth harmonic. Both spectra are two-sided because they involve summations on both sides of $n = 0$. They can be made one-sided by multiplying their values by two and then summing from $n = 0$ to ∞. The power density spectrum is the derivative of the power spectrum. Its ordinate units are those of amplitude squared divided by frequency. So, the integral of the power density spectrum over a particular frequency range yields the power contained in the signal in that range. The spectra of a signal obtained by impulsively tapping a cantilever beam supported on its end, shown in Figure 11.14, displays a dominant frequency at approximately 30 Hz.

11.10 THE CONTINUOUS FOURIER TRANSFORM

Fourier analysis of a periodic signal can be extended to an aperiodic signal by treating the aperiodic signal as a periodic signal with an infinite period. From Equation 11.33, the Fourier trigonometric series representation of a signal with zero mean ($A_0 = 0$) is